Project Activity: Live Airport Flights Dashboard Using Aviationstack API

Activity Title: Live Flights Dashboard for Selected Airports

Objective

In this project, you will build a web page that retrieves **live flight data** from the **Aviationstack API** and displays both **departure flights** and **arrival flights** for a selected airport.

Your application must allow the user to choose an airport from a dropdown list, then load and display:

- **Departure flights**
- **Arrival flights**

Each type must be shown on its **own table**.

Required Airports

Your dropdown list must include these 5 airports:

- **Bahrain Airport**
- **King Fahad Dammam Airport**
- **Riyadh Airport**
- **Cairo Airport**
- **Istanbul Sabiha Airport**

You should use the correct **IATA airport code** for each airport in the dropdown.

Required Layout

You must use **Bootstrap** for styling and page layout.

Layout rules

- Put everything inside a Bootstrap **container**
- Use Bootstrap **grid system**
- Make **two tables**

- The two tables must appear **next to each other**
- Each table must take **6 columns** of the page
- Use a **row** and **two col-6 / equivalent Bootstrap columns**

Required Features

1. Airport selection

Create a **dropdown list** so the user can choose one airport.

2. Load button

Create a button that loads the data for the selected airport.

3. Current date and time

Show the current date and time on the page and keep updating it automatically.

4. Two live data tables

After the user selects an airport and clicks the button, show:

- one table for **Departure Flights**
- one table for **Arrival Flights**

Required Fields to Display

Common flight information

Show these fields in the tables:

- **flight_date**
- **flight_status**
- **airline name**
- **flight iata**

Departure Flights Table

For the **departure** table, show:

- Flight Date
- Flight Status
- Departure Airport
- Departure Terminal
- Departure Delay
- Departure Scheduled
- Departure Estimated
- Departure Actual
- Departure Time Zone
- Airline Name
- Flight IATA

Arrival Flights Table

For the **arrival** table, show:

- Flight Date
- Flight Status
- Arrival Airport
- Arrival Terminal
- Arrival Delay
- Arrival Scheduled
- Arrival Estimated
- Arrival Actual
- Arrival Time Zone
- Airline Name
- Flight IATA

Functional Requirements

Your program must:

- read the selected airport from the dropdown
- send a request to the **Aviationstack API**
- retrieve **departure flights** for the selected airport
- retrieve **arrival flights** for the selected airport
- place the data into two separate HTML table bodies
- handle cases where no data is returned
- update the screen cleanly every time the user chooses another airport

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Important Notes

- Do **not** hardcode flight data manually
- Data must come from the **API**
- You must use **JavaScript + fetch**
- You must use **Bootstrap**
- You must use **separate tables**
- Keep your page organized and readable

Hints Only

Do not copy full code from anywhere. Use these hints to guide your implementation.

HTML structure hints

You will likely need:

- a **select** element for airports
- a **button**
- a label or element to show current time
- two separate **table** elements
- two separate **tbody** elements, one for arrivals and one for departures

Bootstrap hints

You will likely use:

- **container**
- **row**
- **col-6**
- **table**
- **table-bordered**
- **table-dark**
- **form-select**
- **btn btn-primary**

JavaScript function hints

You may create functions with names such as:

- LoadFlights()
- LoadDepartureFlights()
- LoadArrivalFlights()
- Update_Time()
- Set_Time_Interval()
- BuildDepartureTable()
- BuildArrivalTable()

Time update hint

To keep the date/time changing automatically, use:

- **setInterval()**

API request hints

You will likely need:

- **fetch()**
- **await**

- **response.json()**

Filtering hints

To get flights by airport, check the API endpoint parameters for:

- departure airport IATA filter (dep_iata)
- arrival airport IATA filter (arr_iata)

You will probably need **one request for departures** and **one request for arrivals**.

Data access hints

The API response will contain nested objects, so you will need to access properties inside objects such as:

- flight information
- airline information
- departure information
- arrival information

Table generation hints

You may:

- loop through the returned data using a loop such as for
- create rows dynamically
- insert generated rows into the correct tbody using:
    - document.getElementById(...)
    - .innerHTML

Suggested Steps for Students

1. Build the page structure with Bootstrap.
2. Add the airport dropdown with the 5 required airports.
3. Add the current time section.
4. Add two empty tables side by side.
5. Create the time update function.
6. Create the button click function.
7. Send API request for departure flights.
8. Send API request for arrival flights.
9. Fill both tables with the returned results.
10. Test with different airports.

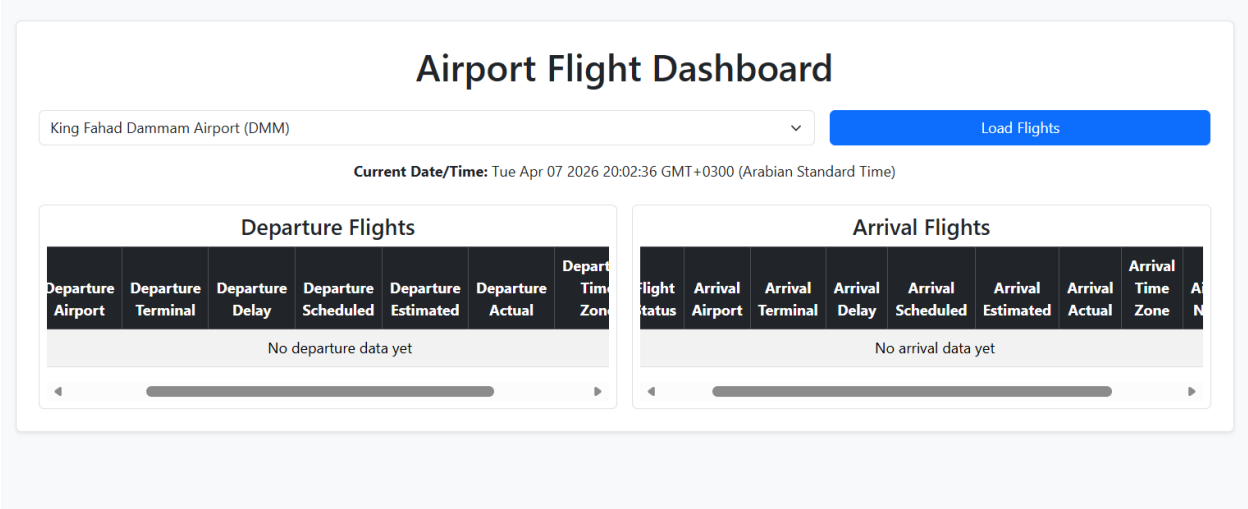
Submission Instructions

Submit the following on the LMS:

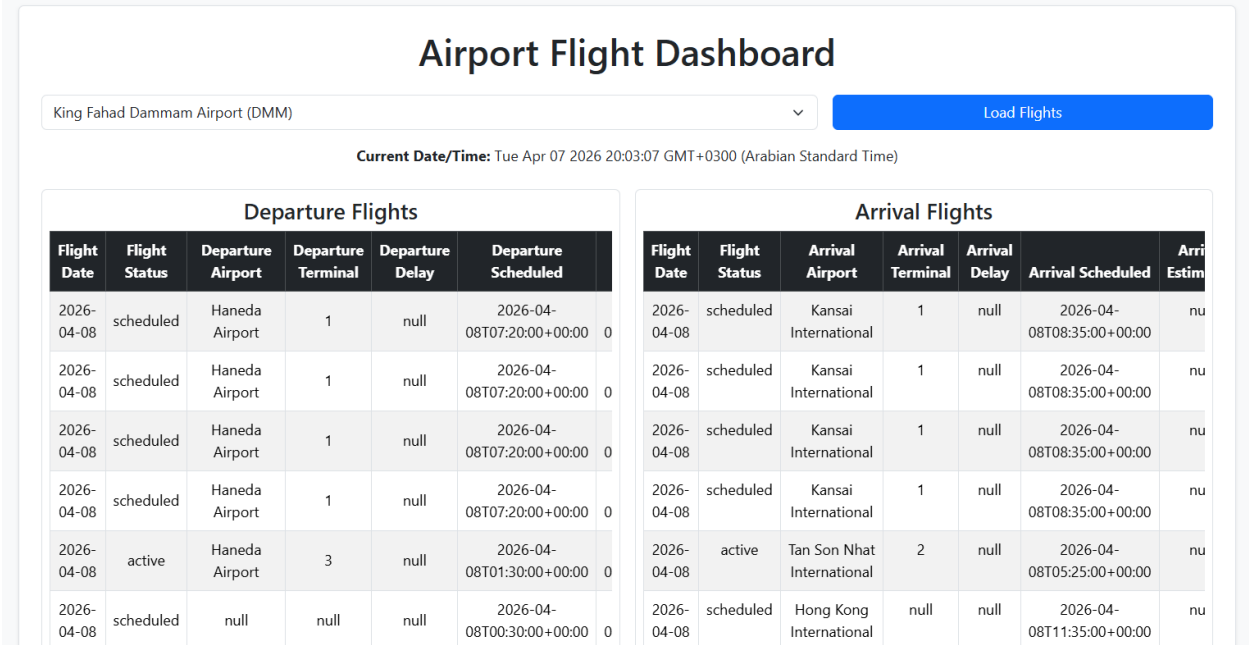
- airport_screen.html

Examples

Example Input:



Example Output



Grading Rubric

You can use this rubric:

- **5%** Bootstrap layout and clean design
- **10%** Correct use of container, row, and col-6
- **10%** Airport dropdown includes all 5 required airports
- **15%** Current date/time updates correctly
- **25%** Departure table works correctly
- **25%** Arrival table works correctly
- **10%** Correct fields displayed from API

Extra Challenge

For bonus marks, students may add:

- loading message while fetching data
- error message if API fails
- refresh button